Experiment 2: Preparation of a Software Project Plan Covering Scope, Timeline and Team Roles

Aim

To prepare a software project plan that defines the scope, timeline and team roles for a sample application.

Objective

- Define the scope and boundaries of the project.
- Break the work into phases with a realistic timeline.
- Assign clear roles and responsibilities to team members.
- Identify risks and plan how to handle them.

Selected Application

Online Library Management System - a web application that lets students search for books, reserve or borrow them, and lets librarians manage the catalogue and fines.

1. Project Scope

In scope

- Student and librarian login with role-based access.
- Book catalogue: add, edit, delete and search books.
- Issue, return and renewal of books.
- Automatic fine calculation for late returns.
- Monthly reports on issued books and fines.

Out of scope

- Online payment of fines (planned for a later release).
- Mobile application.
- E-book reading inside the system.

Assumptions and constraints

- Existing book data is available in Excel format.
- Team size is 5 members; development period is 10 weeks.
- Hosting on a free-tier cloud service.

2. Project Timeline

Phase	Activities	Duration	Weeks
Requirement analysis	Meetings, SRS preparation, sign-off	1 week	1
Design	Database schema, UI wireframes, architecture	2 weeks	2 - 3
Development	Login, catalogue, issue/return, fines, reports	4 weeks	4 - 7
Testing	Unit, integration, system testing, bug fixing	2 weeks	8 - 9

Phase	Activities	Duration	Weeks
Deployment	Hosting, user training, hand-over	1 week	10

3. Team Roles and Responsibilities

Role	Responsibility
Project Manager	Planning, tracking progress, communication with the client, risk management
Business Analyst	Collecting requirements and preparing the SRS
UI/UX Designer	Wireframes, screen design, usability
Developer(s)	Front-end and back-end coding, database implementation
Tester (QA)	Writing test cases, executing tests, reporting defects

4. Deliverables

- Software Requirement Specification (SRS)
- Design document and database schema
- Working application (source code)
- Test report
- User manual

5. Risk Management

Risk	Impact	Mitigation
Requirements change late	High	Fix scope after sign-off; handle changes through change requests
Team member unavailable	Medium	Share knowledge, keep documentation updated
Schedule delay	Medium	Weekly review, buffer time in testing phase
Data migration errors	Medium	Validate Excel data before import

Results

A complete project plan was prepared with clear scope, a 10-week schedule, defined roles and a risk register. The plan gives the team a common reference for tracking progress.

Conclusion

A good project plan reduces confusion, controls cost and time, and makes responsibilities clear. Defining the scope early and reviewing progress every week helps the project to be delivered on time.